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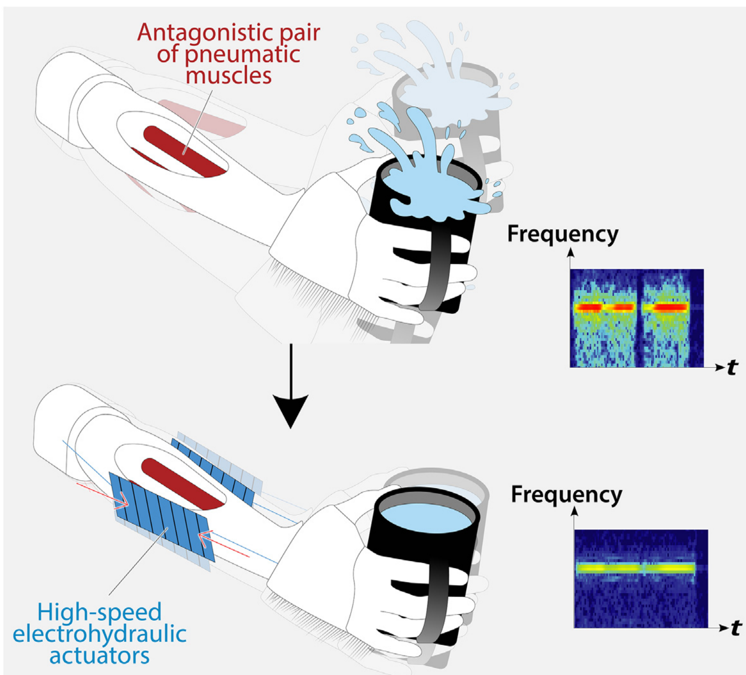
A robotic and virtual testing platform highlighting the promise of soft wearable actuators for wrist tremor suppression

Graphical abstract



Highlights

- Reproduction of patient recordings of tremor episodes in a robotic platform
- Slim and lightweight Peano-HASEL actuators tested for suppressing tremors
- Biomechanical model validates that forces are sufficient for practical applications
- Evaluation tools reduce time and resource requirements for testing prototypes

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In brief

Shagan Shomron et al. created a robotic and simulation testing platform for wearable tremor-suppression devices. The robotic test bed, or "mechanical patient," can reproduce recordings of tremor episodes and be used for optimizing early-stage designs instead of time-consuming clinical trials. Electrohydraulic actuators tested on the platform were shown to be promising for the suppression of human wrist tremors.

4 **Validate**
Functional device with real-world testing, ready to scale

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A robotic and virtual testing platform highlighting the promise of soft wearable actuators for wrist tremor suppression

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THE BIGGER PICTURE Millions of people worldwide live with tremors, involuntary oscillating movements caused by diseases such as Parkinson's, which disrupt their ability to perform everyday tasks, such as handling a cup of coffee. Traditional treatments, including medications and surgeries, may display significant side effects. Wearable soft robotic devices may offer non-invasive and customizable solutions for actively suppressing tremors. However, from the perspective of product development, early clinical testing of such technologies can be very time consuming and resource intensive. To help relieve this developmental bottleneck, we created a "mechanical patient" that mimics real tremor episodes to test the capability of soft robotic actuators to suppress tremors. Using the mechanical patient alongside simulations, we demonstrate that a pair of slim, lightweight, silent, and soft electrohydraulic actuators are fast and strong enough to suppress clinically relevant tremors in the wrist.

SUMMARY

Nearly 80 million people in the world deal with medical conditions that cause involuntary periodic movements known as tremors. Wearable soft robotic devices offer a potential solution for actively suppressing these tremors. However, existing prototypes face limitations in actuation performance and complex testing procedures. We present a comprehensive approach for the rapid evaluation of emerging wearable tremor-suppression technologies. This method combines reproducing patient-recorded tremor episodes and measuring tremor suppression in a robotic platform, termed a "mechanical patient", with validation of the achieved suppression performance of soft actuators via biomechanical modeling, thereby avoiding time-consuming clinical testing in the early stages of development. Using this approach, we highlight that an antagonistic pair of slim and lightweight electrohydraulic actuators can effectively suppress clinically relevant tremors between 2 and 8 Hz. The biomechanical model confirms that the pair of actuators generates adequate suppression forces for all tested tremors.

INTRODUCTION

Tremor affects an estimated 80 million individuals worldwide and is a characteristic and debilitating symptom of Parkinson's disease (PD), essential tremor (ET), and other neurological diseases.[1,2] Many of those who live with upper limb tremors struggle to perform activities of daily living, such as drinking from a glass or writing, resulting in a decreased quality of life.[3] Current medical-symptomatic treatments for tremors span from prescription drugs to stereotactic surgical intervention.[4–6] Although these established treatments show relevant symptomatic improvement, limitations remain. For instance, the tremor may worsen upon disease progression, invasive surgery poses other health risks, and tolerance effects can diminish the efficacy of deep brain stimulation in the long term.[7] As a result, some of these options are undesirable or incompatible for some patients.[5,8–10]

In recent years, extensive research has focused on developing wearable orthoses for tremor suppression to overcome these issues with non-pharmacological, non-interventional strategies.[11] To be accepted by patients and amenable for everyday use outside of the clinic, wearable assistive devices must satisfy several requirements. They should be comfortable to wear, operate quietly, and be aesthetically pleasing so as to not draw unwanted attention.[12,13] Additionally, the devices should be easily controlled to autonomously and quickly react to and suppress tremorous movements but not obstruct the intended movements.[11,14] While currently available wearable assistive devices are effective in tremor reduction, none of them fulfill all essential criteria, as their bulk, weight, and rigidity pose challenges for long-term usage in patients.[15–17]

Made of soft materials with similar mechanical properties to human tissues, wearable soft robotic devices have inherent advantages over their rigid counterparts,[18–20] and researchers have developed devices and actuators for suppressing tremors.[21–26] For example, Mohammadi et al. developed a phase transition artificial muscle operated by an energy-efficient reinforcement learning controller that could passively damp tremor movements and vibrations[27,28]; Wirekoh et al. developed a fiber-reinforced bending pneumatic actuator, combined into a finger-wearable device, that can operate within the frequency and amplitude range of human tremors[16]; and Zahedi et al. developed a rotational semi-active actuator based on magnetorheological fluid to suppress elbow tremors.[15] However, most of these mechanisms still face limitations, such as bulky and noisy components that power pneumatic devices and tether them to a power source or stiffening of semi-active devices that restrain the intentional and natural movements of the patient. In contrast, electrically driven soft actuators, such as dielectric elastomer actuators,[29] have potential in tremor-suppression applications, as they are energy efficient and fast.[30] Among the electrically driven soft actuators reviewed by Fromme et al., Peano-HASELs, a biomimetic form of hydraulically amplified self-healing electrostatic (HASEL) actuators,[31,32] were particularly promising candidates for tremor suppression.[11] Peano-HASELs linearly contract using high electric fields to locally displace liquid dielectrics enclosed in soft electrohydraulic architectures.[33–35] These slim and lightweight actuators feature muscle-like performance, high bandwidth (>30 Hz), and silent operation while showing high controllability and stable contraction over time and having a low material cost.[36,37]

However, it is challenging to evaluate the performance of newly developed actuators under realistic clinical conditions, partially because the technology may not be ready to be embedded into a device that meets medical safety regulations for clinical trials,[38,39] and so far, there is only one publication where a wearable soft robotic device was tested on a single human patient,[23] while some other studies have been tested in a lab setting on healthy individuals[40] who voluntarily oscillate their hands or wear gloves that simulate a tremor with electrical muscle stimulation.[17,41] While computational models can, in principle, be used to predict the required assistive forces,[42,43] these models also face challenges to accurately describe soft actuator dynamics. Thus, simulation alone is not sufficient for extensive evaluation of novel actuation approaches. Alternatively, combining neuromechanical simulations[44–48] with a biomimetic test bed,[49,50] to form a "mechanical patient" for laboratory use, can be a beneficial approach to evaluate novel actuator technologies.[51]

Here, we present a comprehensive approach for the evaluation and validation of soft robotic technologies for active suppression of human wrist tremors. This method combines the reproduction of patient recordings of tremor episodes in a robotic test bed, i.e., a mechanical patient, that mimics the human forearm, with validation of the forces required for the suppression of tremors via biomechanical computer simulation. We then use this approach to highlight the potential of an antagonistic pair of Peano-HASEL actuators in suppressing human tremor trajectories that were recorded from patients with different disease etiologies. We reproduce tremors with a mechanical patient that uses pneumatic artificial muscles (PAMs) that are operated antagonistically to mimic the human forearm (Figure 1) and suppress them using a single slim (∼1 mm) and lightweight (∼15 g) Peano-HASEL actuator on each side of the setup. Our results show a drastic reduction of tremor amplitude ranging between 76% and 94% over a wide range of clinically observed tremor severities and frequencies between 2 and 8 Hz, spanning more than the typical range of frequencies seen in patients with PD[52] and ET.[53] Despite their non-linearity, tailoring the response of Peano-HASEL actuators with a simple proportional-integral-derivative (PID) controller enables a reduction of more complex patient-recorded tremor episodes by at least one grade on a clinically validated tremor rating scale. The biomechanical simulation results confirm that sufficient suppressive forces are achieved using this single antagonistic pair of Peano-HASEL actuators. This comprehensive approach helps avoid time-consuming clinical testing in the early stages of device development and highlights the potential of these soft electrically driven actuators to be implemented and improve the quality of life of many patients.

RESULTS

Design and control of the mechanical patient

The mechanical patient is a robotic test bed that we designed and fabricated to mimic the human forearm, wrist, and hand (Figures 2 and S1). The system consists of a three-dimensional (3D)-printed forearm, with a rotating-3D-printed hand (Figure 2A) that weighs 450 g and is shown holding a mug for tremor visualization (Video S1). We installed two McKibben PAMs inside the 3D-printed forearm (Figures 2B and S2). By alternating inflation and deflation, the PAMs function as an antagonistic pair of muscles, generating flexion and extension movements about the wrist joint (Figure 2D). We chose to use PAMs rather than direct current (DC) motors because textile-braided PAMs exhibit viscoelastic force

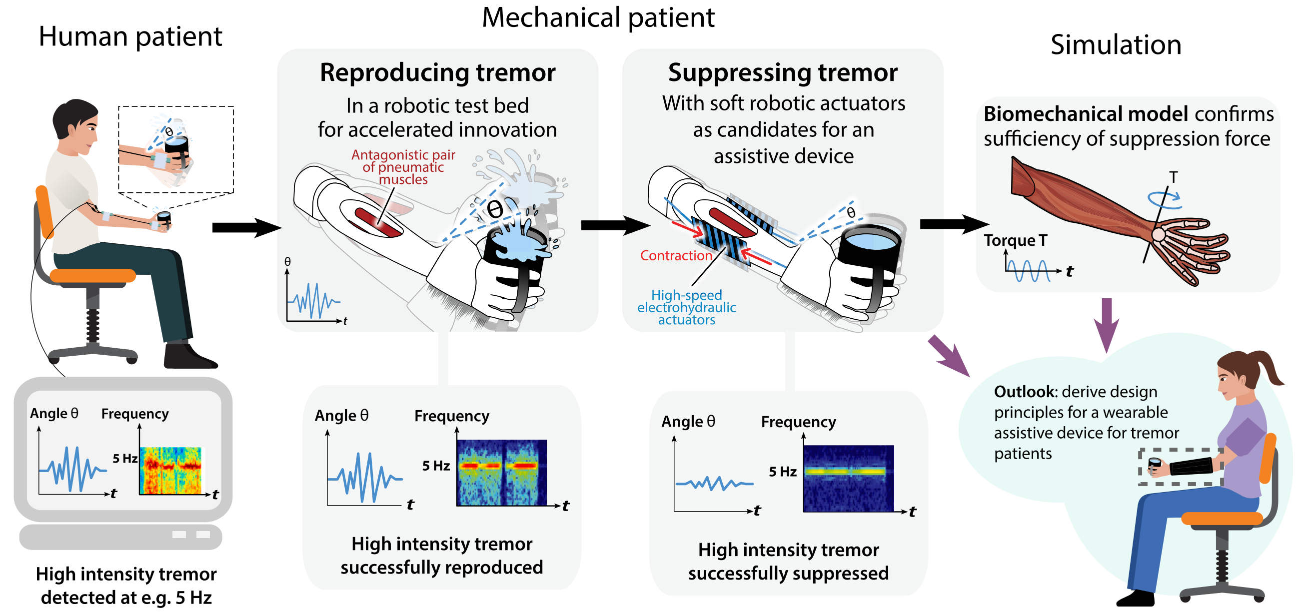

Figure 1. Comprehensive approach for a rapid evaluation of emerging tremor-suppression technologies
The process starts in the clinic, where patient data for the characteristic frequencies and amplitudes of tremors are collected. The tremor is then reproduced in a mechanical patient, a biomimetic robotic test bed based on an antagonistic pair of pneumatic muscles. Suppression of the tremor is then achieved using an antagonistic pair of high-speed electrohydraulic actuators that are controlled to counteract the wrist tremor. Next, a biomechanical model of wrist tremors is used to confirm the sufficiency of suppressive forces. Overall, this approach serves to evaluate emerging technologies for use in wearable assistive devices while avoiding time-consuming clinical testing in early stages of development.

dynamics similar to biological skeletal muscles[54]; hence, they are suitable for replicating natural muscle movements.[49,55,56] Compared with biological muscles, PAMs have similar maximum actuation stress, actuation response, and load curves.[57,58] Moreover, by initially investigating 1 degree of freedom (DOF) prior to extending to multiple ones, the mechanical patient approach allows for a thorough investigation of system components and suppression technologies.

We controlled the system as hardware in the loop via MATLAB Simulink Real-Time Desktop (Figure 2E). To represent typical wrist tremors in the mechanical patient, we utilized sine-wave control signals, controlling the pressure amplitude, frequency, and co-contraction values of the PAMs. In order to reproduce an equivalent to human tremors in the mechanical patient, specific parameters were chosen in accordance with the literature,[59] as well as data collected from the volunteers participating in the study presented in this paper. The pneumatic muscles allow for an active range of motion of 70° about the wrist. This range is slightly lower than the range of voluntary motion in activities of daily living (about 80°[60]), but it was sufficient to test the whole range of tremors, given that the tremor is generated with a mean angle of 0° and typically has a range of less than 12°. Other joint angles could be investigated with the same setup after minor adjustments to the housing of the pneumatic muscles. Although the mechanical patient setup does not consider the detailed non-linear elastic response of deformable skin, which may affect the performance of a wearable soft exoskeleton, we included two steel springs in series with the actuators to approximate skin deformation close to the attachment points.

In the following sections, we first describe how we utilized the mechanical patient and the Peano-HASELs for the generation and suppression of typical sinusoidal tremors for a systematic grid of different fixed amplitudes and frequencies spanning a wide range of tremors associated with different disease etiologies according to the literature. Subsequently, we replicate clinically measured episodes of patients with tremors diagnosed with PD or ET (Table S1) and demonstrate that a simple control approach is sufficient to suppress specific patient patterns by adapting to the varying amplitudes.

We used the mechanical patient to represent typical human tremors according to the literature.[59] A spectral analysis of the episodes that we collected from our patients showed a distinct peak for every patient indicating a predominately sinusoidal movement of the wrist during tremor episodes (Figure S3). We generated a tremor in the mechanical patient based on sinusoidal control signals. We classified the tremor amplitude A as the length of the arc the fingertip traces between the greatest points of flexion and extension of the wrist (Figure 2D) in accordance with clinical tremor rating. We rate tremor severity on a custom scale (Table S2) spanning clinically validated tremor rating scales for PD and ET.[61,62] The amplitude was adjusted between 0.7 and 4 cm, in accordance with four different tremor grades (Table 1), between a barely perceivable and a severe tremor.[63,64] To demonstrate the capabilities of the mechanical patient, we chose frequencies in the range of 2–8 Hz, which covers slightly more than the characteristic range of frequencies of PD and ET.[59,65]

By adjusting the sinusoidal control parameters for the pressure to the PAMs, we were able to reach each desired tremor

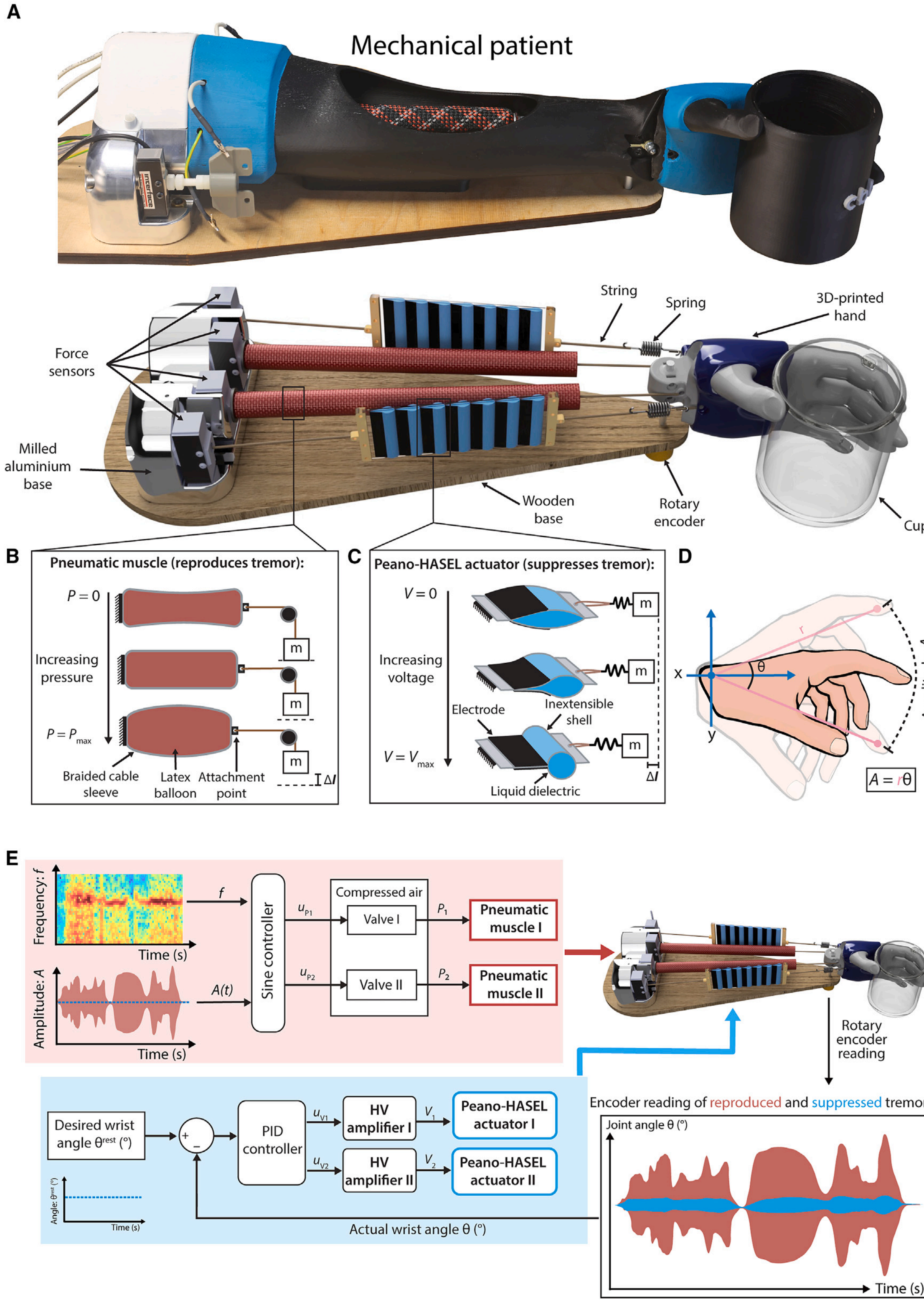

Figure 2. Design and control of the mechanical patient and tremor-suppression mechanism

(A) Photograph (top) and schematic design (bottom) of the mechanical patient, a biomimetic robotic test bed, that is composed of forearm and hand and operated by an antagonistic pair of pneumatic artificial muscles (PAMs) that controls the movement of the wrist and generates the tremor. The schematic design also shows an antagonistic pair of Peano-HASEL actuators superimposed on the dorsal and ventral sides of the arm and used to suppress the generated tremor movements.

(legend continued on next page)

Table 1. Custom scale of tremor rating grades

Tremor grade	Amplitude (cm)	θ (°)
Grade 1 (barely perceivable)	<0.7	<2
Grade 2 (moderate)	0.7–2	2–5.7
Grade 3 (marked)	2–4	5.7–11.5
Grade 4 (severe)	>4	>11.5

Each grade is represented by a description, an amplitude in cm, and an angle (θ) in degrees, which corresponds to the reading from the encoder and the values from the simulation.

grade and frequency and recorded the position of the wrist over a few minutes. The results show that our biomimetic system stably generates a tremor in the desired frequency ranges and is adjustable to different tremor grades (Figure S4). It is important to note that, as seen in the clinic, at higher frequencies, the maximum possible tremor grade reduces, indicating an inverse relationship between the maximum tremor amplitude and frequency.[66] As such, all four grades were generated at lower frequencies of 2–4 Hz, and only the lower grades were generated for frequencies above 6 Hz.

Evaluating tremor-suppression performance using the mechanical patient

HASEL actuators have been explored for a range of robotics applications[67–69]; here, we employ a subclass named Peano-HASEL actuators,[32] which linearly contract upon activation with voltage and can directly counteract the antagonistic pair of PAMs (Figures 2C and S5). In this study, the actuators were fabricated to fit along an average human male forearm from the elbow to the wrist, but they may be easily scaled up or down to match different arm sizes due to their simple and flexible fabrication method. To suppress the tremor generated by the PAMs, we superimposed an antagonistic pair of Peano-HASEL actuators, one on the ventral side and one on the dorsal side of the mechanical patient (Figure 2A), and repeated the procedure described above to generate sine-wave signals with human tremor characteristics in the mechanical patient.

To achieve suppression of typical sinusoidal tremors, we controlled the driving voltage to the Peano-HASEL actuators using a PID controller that was programmed to minimize the joint angle (Figure 2E). To quantify the effectiveness of tremor suppression, we recorded the tremor amplitude with the tremor-suppression mechanism activated and then deactivated for intervals of 20 s (Figure 3). For each set of frequency and tremor rating grade, we repeated the activation and deactivation 6 times to ensure repeatability (Figure S6), and we demonstrated the system's stability over 30 min for a 2-Hz, grade 2 tremor (Figure S7). Our results show the highly successful suppression of tremor amplitudes (76%–93%) across all frequencies between 2 and 8 Hz (Figure 3).

To demonstrate the reproduction and suppression of realistic tremor episodes, we collected wrist angle data from several individual patients (Figure S8) and chose four (Table S1) who displayed distinct tremor episodes with clear onsets, variations, and decay within 2–3 min. We used both the normalized trajectories and the dominant frequency, obtained by a Fourier transform, to reproduce tremor episodes in the mechanical patient (Figure 4). To suppress the tremor, we used the same PID controller for the Peano-HASEL actuators and measured the suppressed amplitude (Figures 4E, 4F, and S6). A spectrogram of the amplitude shows that for patient 4, the magnitude of the suppressed episode is about 10 dB lower than its unsuppressed counterpart (Figures 4H and 4I). Similarly, we achieved a reduction of 10–15 dB with data that were obtained from three other patients (Figure S8), which corresponds to a reduction of at least one tremor grade.

At the moment, neither the tremor monitoring and recording approach nor the tremor-suppression control is capable of distinguishing between intended voluntary movement and involuntary tremor. We performed offline high-pass filtering to remove the voluntary movements measured from patients and assumed a desired joint angle of 0° in the control loop. While this simplification was sufficient for testing the capabilities of soft actuators, enhancing the mechanical patient and this simulation to also account for voluntary movement would be an important step toward the development of effective assistive devices and would make the approach relevant for testing more advanced sensing and control concepts.[70]

Validating tremor-suppression forces using biomechanical modeling

We utilized a biomechanical model to estimate if the forces generated by the suppression mechanism would suffice to suppress tremors in a real patient instead of the mechanical patient.[42] Specifically, we performed forward-dynamic simulations with an OpenSim[71] musculo-skeletal model derived from McFarland et al. and Saul et al.,[72,73] which considers the contribution of muscular viscoelasticity in estimating the required assistive forces.[42] We also generated a tremor with sinusoidal stimulation signals to the wrist flexor and extensor muscles. Additionally, we applied an open-loop sinusoidal torque directly to the wrist joint and tuned its amplitude and phase shift to maximally suppress the tremor amplitude. The tuned torque amplitude was the basis for our estimate of the required assistive force.

To compare the required assistive forces from our simulation results to those that the Peano-HASELs generated during contraction, we recorded the forces with tensile load cells

(B) Actuation mechanism of PAMs, which inflate as air pressure increases, leading to a linear contraction upon the activation axis.
(C) Actuation mechanism of Peano-HASEL actuators, which linearly contract upon activation with voltage by displacing liquid dielectric.
(D) Definition of tremor amplitude, measured by the finger position along the arc, where *A* is the amplitude, *r* is the radius, and θ is the angle between two radii along the arc.
(E) Schematic representation of the control environment for the pneumatic muscles generating the tremor (red) and the soft actuators suppressing the tremor (blue).

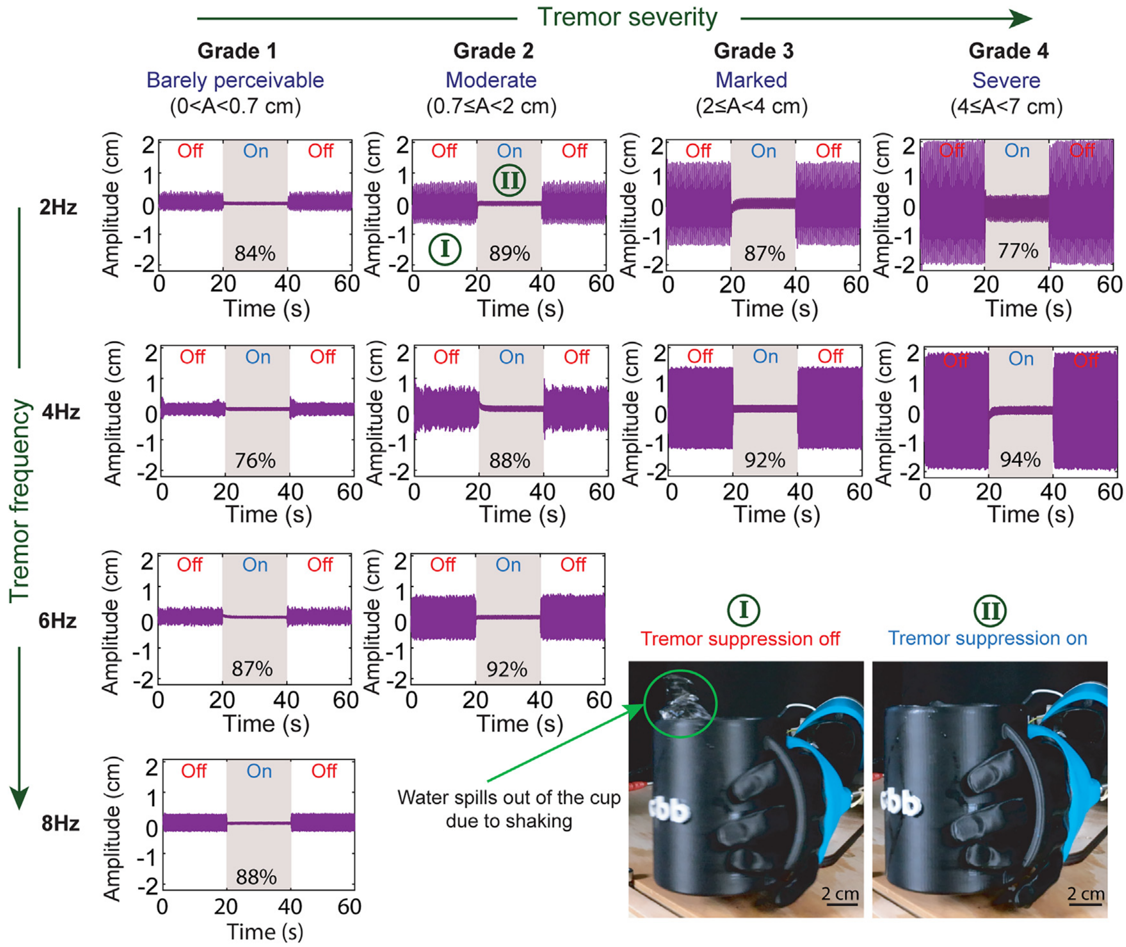

Figure 3. Suppression of typical sinusoidal tremors of varying frequency and severity

Sine-wave signals representing different tremor rating scales (grades 1–4, corresponding to mild to severe) at different frequencies (2–8 Hz) were generated and suppressed in the mechanical patient. The suppression performance is presented as the percentage of amplitude reduction between the activated (on) and deactivated (off) states of the tremor-suppression mechanism. A mug filled with water was used only to visualize successful tremor suppression (still images taken from Video S1). Scale bar, 2 cm.

attached to the mechanical patient, one for each actuator (Figure 2A). These forces were recorded during the experiments described above. Our results suggest that at all measured conditions, Peano-HASEL actuators could generate higher dynamic sinusoidal forces than those predicted by the simulation as minimum required suppressive forces (Figure 5).

We further measured the maximum force that a Peano-HASEL actuator can exert at zero displacement (blocking force, Figure S9). The actuator was mounted horizontally to align with the configuration of the mechanical patient. We found that even a single actuator reaches a blocking force of 40 N, which is well above the required suppression forces predicted by the simulation, suggesting that an even smaller and more lightweight pair of Peano-HASELs would be sufficient.

DISCUSSION

In this study, we have presented a comprehensive approach for assessing the performance of tremor-suppression technologies while highlighting Peano-HASEL actuators as promising candidates for active suppression devices. We showed that one pair of slim (∼1 mm) and lightweight (∼15 g) Peano-HASELs is sufficiently strong and fast enough to reduce tremor trajectories within a broad range of frequencies (2–8 Hz), which include those exhibited in patients with PD and ET. We showcased that developing mechanical patients could enable the evaluation of different technologies in the early stages of research, saving time, effort, and funds prior to clinical trials.

A few additional steps have to be tackled before Peano-HASEL actuators can be used for tremor suppression in clinical applications. For example, the suggested configuration of the actuators was designed to suppress tremors in only one reference joint angle; hence, adding a clutch mechanism could allow for tremor suppression in other wrist positions. In addition, to ensure electrical safety, sufficient insulation and a safe level of maximum discharge currents must be considered. These requirements can be met with the proposed size of actuators[74,75] and integrated with portable and safe electronic circuits that are

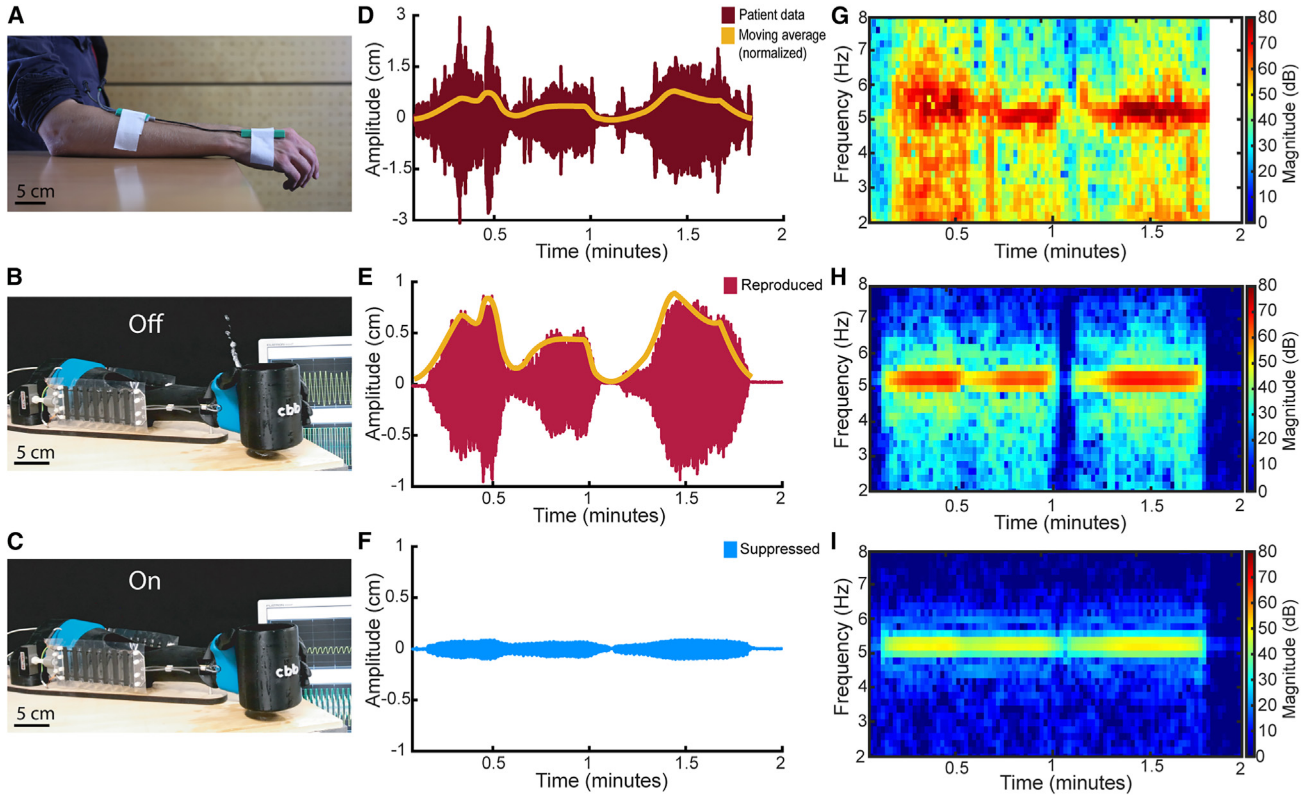

Figure 4. Suppression of patient recordings of tremor episodes

(A) A representative photo of the measuring device mounted on a person. Scale bar, 5 cm.

(B and C) A representative photo of the mechanical patient in deactivated (B) and activated (C) modes of the suppression mechanism. Scale bar, 5 cm.

(D) Normalized recorded amplitude over time with its moving average.

(E and F) Reproduction (E) and suppression (F) of measured tremor episode from patient 4, who is diagnosed with essential tremor, during a tremor episode (see Table S1 for details). The tremor was measured in a holding condition while the patient was challenged with cognitive tasks.

(G–I) Short-time Fourier transform of the tremor episode as recorded from patient 4 (G), as reproduced in the mechanical patient (H), and as suppressed using electrohydraulic actuation (I).

already being developed.[76] In addition, the mechanical patient in this study considers only 1 DOF and was using PAMs, where dynamic materials behavior is difficult to model. Extending the concept to multiple joints would make the system more complex, but it would also enhance the mechanical patient's similarity to a human patient. This would allow studying superimposed frequencies at the wrist, which may result from interaction torques and biarticular muscles[77] driven by tremors of varying amplitudes[78] or frequencies[79] across different joints. Additionally, the neuronal control of the mechanical patient and the biomechanical model was simplified to a sinusoidal feedforward signal close to minimal muscle activation, which neglects the possible contribution of natural reflexes or additional neurological symptoms, like increased muscle tone, that may affect external tremor suppression in humans. Such contributions to tremors could be examined in further refined mechanical patients and models considering multiple joints, biarticular muscles, and refined neuro-muscular control schemes.

The evaluation platform proposed in this study can be extended to study other movement disorders, such as dysmetria in cerebellar ataxia[42] or spasticity resulting from hypersensitive reflexes, by generating PAM control signals that mimic these conditions.[80] Here, a simple PID controller is used to evaluate the basic capabilities of the actuators, such as actuation forces and reaction speeds. To improve the reaction and suppression rates and account for voluntary movements, advanced control approaches that utilize Kalman filters,[70] admittance, or model-based control[81–84] could be considered. Future research efforts could also follow new advances in the field of user intention detection[85–89] and context awareness.[90]

METHODS

Collection of data from patients with tremor

We aimed at recording exemplary tremor episodes, which show clear onsets, some amplitude variation, and decay, for reproduction in the mechanical patient. To this end, we measured the wrist angle of 10 patients while evoking tremors. We finally selected single episodes from four patients, three males and one female, with a mean age of 76 ± 3.8 years (Table S1). Two patients were diagnosed with ET and two with PD. All subjects gave their informed consent for inclusion before they

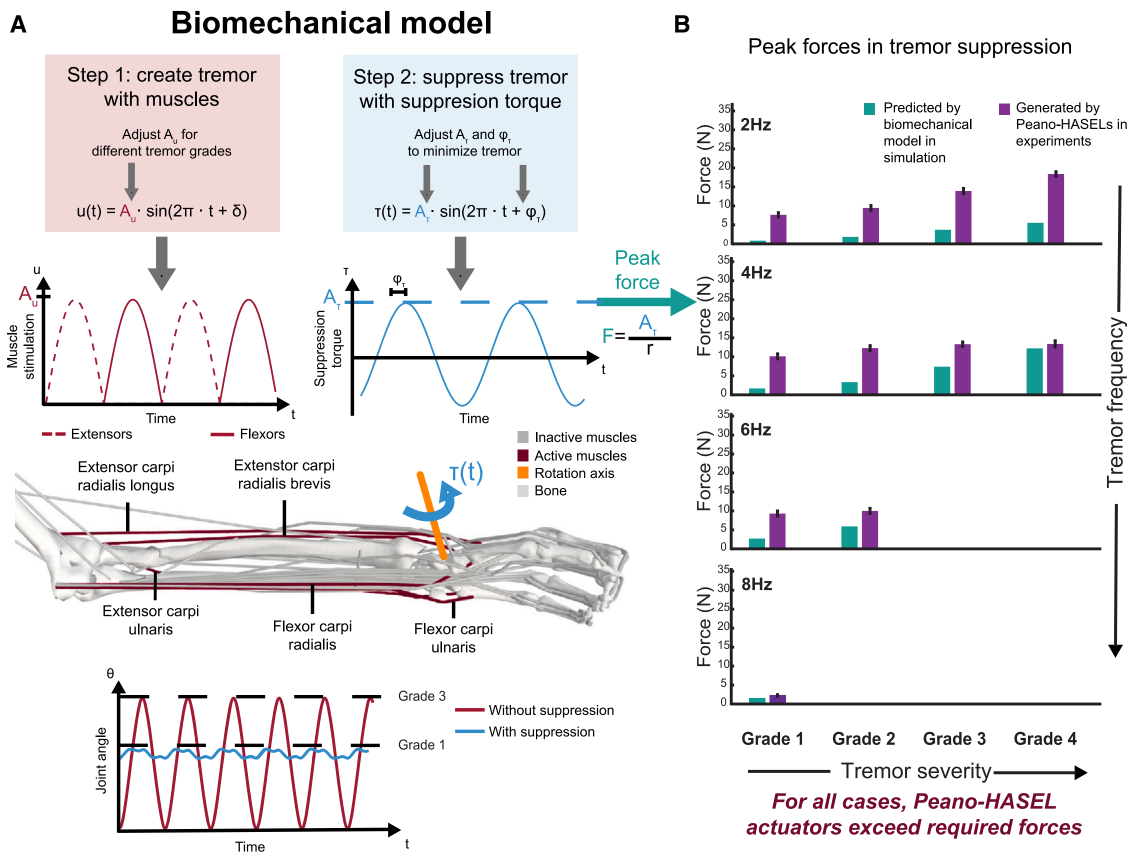

Figure 5. Biomechanical modeling to confirm sufficiency of suppression forces

(A) Depiction of the biomechanical model, including equations for creating and suppressing the tremor, a graphic description of extracting the peak force, and a schematic illustration of the modeled wrist.

(B) Forces required for suppression of tremor at different tremor frequencies and severities as predicted by the simulation (green). Experimentally measured forces from a single pair of Peano-HASEL actuators (purple) exceed the required forces across all frequencies and severities of tremor. The results represent the mean and standard deviation of 6 repetitions.

participated in the study. The study was conducted in accordance with the Declaration of Helsinki and the Art. 6 Abs. 1 lit.a and Art. 9 Abs. 2 lit.a of the Datenschutz-Grundverordnung (DSGVO). The protocol was approved by the Ethics Committee of the University Hospital Tübingen/Germany (961/2021BO1). At the time of data collection, the patients were on their respective medications, allowing for representations of tremors with medications.

For data collection, a dual-axis goniometer (TSD130A, BIOPAC Systems, CA, USA) was attached to the forearm and the back of the dominant hand (Figure 4A). We recorded angular deflection of the wrist about the flexion/extension and radial/ulnar axes. As our aim was to record episodes of strong tremors, patients were challenged with cognitive tasks while their hands and arms were either relaxed in their lap or extended straight in front of their body, which we named resting and holding conditions, respectively. From the collected data, we extracted episodes of tremors, for which we determined the mean amplitude and the frequency using short-time Fourier transformation. The episodes we chose for reproduction in the mechanical patient represent individual tremor episodes from 4 patients with clear onsets, tremor variations, and decay with durations between 2 and 3 min.

Design and fabrication of the mechanical patient

The mechanical patient consists of two 3D-printed segments of a human-sized forearm and hand (filament: ABS, Stratasys Fortus 450 mc, MN, USA) connected by one hinge joint allowing for flexion-extension of the wrist. Two PAMs were responsible for generating wrist movement and were fabricated from an inner latex tube with a diameter of 10 mm and a thickness of 0.3 mm and an outer braided textile hose with a diameter of 13–26 mm (VIABLUETM, Germany). The latex tube and braided hose were fastened at both ends with 16-mm-long M6 screws, 8.4-mm washers (DIN125), and M6 nuts or cap nuts (Figure S2). Kevlar strings were used to mount the PAMs to the wrist of the mechanical patient (Figures 2A and S1), where they were attached to a 25-mm-diameter pulley. To control the pressure and thus the contraction of the pneumatic muscles, we used two linear pneumatic valves (VPPX, Festo, Germany). We recorded the wrist angle with an encoder (2RMHF5000-D, 5000 ppr., Scancon, Denmark) and the forces of the two PAMs with two strain-gage sensors (model: SML-110N, interfaceforce e.K., Germany). Two additional strain-gage sensors were attached to the outer side of the 3D-printed forearm, where soft actuators for tremor suppression could be mounted and monitored as well. For easy mounting of these actuators, a custom-3D-printed mount was connected directly to each one of these outer sensors via an M6 threaded nylon rod.

Design and fabrication of Peano-HASEL actuators

The fabrication method of Peano-HASEL actuators was previously described in detail.[32,36] Briefly, the actuators were fabricated by heat sealing two sheets of 20-μm-thick weld-seal-ovenable polyester foil (L0WS, multiPlastic, UK) into a 7-pouch geometry. Each one of the 7 pouches was 60 mm wide and 20 mm long. Then, we covered the outer side of the foil with conductive carbon ink (CI-2051, Nagase ChemteX America, OH, USA). Each pouch was then filled with 1.7 mL of 5-cSt silicone oil (Carl Roth, Germany), resulting in an ~1-mm-thick actuator that weighed less than 15 g. Two 1.5-mm-thick acrylic mounts were attached to the outer side of both ends of each actuator to ensure easy and safe mounting onto the mechanical patient. The proximal ends of the Peano-HASEL actuators were attached via M4 nylon screws to the custom-3D-printed mount, while the distal ends were connected to a 50-mm aluminum M4 screw, attached at the wrist, via a Kevlar string and a spring (stainless steel, 1600 N m^{-1}). To operate the Peano-HASELs, we used 2 high-voltage amplifiers (Trek 610E, Advanced Energy, CO, US), one for each actuator.

Control of the mechanical patient and soft actuators

A simplified diagram of the control environment is shown in Figure 2E. We implemented the control as well as the recording of the movement in a hardware in-the-loop setup based on MATLAB/Simulink Real-Time Desktop and a Humusoft MF634 data acquisition card at 2-kHz update frequency. The signals encoding the desired pressure for the PAMs were given as analog output signals to two linear pneumatic valves, which regulated the pressure of the PAMs in the range of 0–2 bar. The signals encoding the desired voltage of the Peano-HASELs were given as analog signals to two high-voltage amplifiers to apply the operating voltages in the range of 0–8 kV. The data acquisition card recorded the signals from the incremental rotary encoder and the amplifiers of the force sensors (DM1, LEG Industrie-Elektronik, Germany).

We controlled the PAMs by applying the following antagonistic sine-wave encodings for the desired PAM pressure values in bar:

$$u_{P1}(t) = P(t)\cdot\sin(f(t)\cdot t) + C_1 \qquad \text{(Equation 1)}$$

$$u_{P2}(t) = P(t)\cdot\sin(f(t)\cdot t + \pi) + C_2 \qquad \text{(Equation 2)}$$

Here, $u_{p1,2}(t)$ are the desired PAM pressures as inputs to the valves. $P(t)$ and $f(t)$ are the amplitude and frequency of the applied sine-waves, respectively. C_1 and C_2 are offset parameters for the two PAMs, which allow us to adjust the rest angle θ^{rest} of the wrist, with respect to flexion and extension. In order to conveniently adjust θ^{rest}, we have chosen the parameters $C_{1,2}$ according to

$$C_{1/2} = B_p \pm D_p \qquad \text{(Equation 3)}$$

with the inputs of B_p, a base contraction level tuned to avoid slacking of the PAMs, and D_p, a distributive pressure offset between the two PAMs to adjust the rest angle θ^{rest} of the wrist.

For the experiments where we generated and suppressed typical sinusoidal tremors, we used a range of constant amplitudes P and frequencies f. To replicate the tremor patient episodes, we used time-varying amplitudes $P(t)$ and constant patient-specific frequencies f.

To operate the Peano-HASELs and suppress the PAM-induced tremor, we applied the following closed-loop PID controller, a suitable control strategy for high-bandwidth active tremor suppression[91]:

$$u_{H1/2}(t) = B_H \pm \mathrm{PID}(\theta^{err}(t)) \qquad \text{(Equation 4)}$$

where u_H is the input voltage for the Peano-HASELs and $\theta^{err}(t) = \theta^{rest}\text{-}\theta(t)$ is the control error of the wrist angle $\theta(t)$ that is recorded in real time by the rotary encoder. The PID control function $\mathrm{PID}(\theta^{err}(t))$ follows a built-in MATLAB/Simulink function block of an ideal PID controller with the transfer function

$$\mathrm{PID}(s) = K_P(1 + K_I \cdot 1/s + K_D \cdot N_{PID}/(1 + N_{PID} 1/s)) \qquad \text{(Equation 5)}$$

with K_P, K_I, and K_D being the heuristically determined controller parameters and $N_{PID} = 100$ the filter coefficient.

Note that the calculation of the PID signal (Equation 5) is the same for both Peano-HASEL actuators. For one actuator, it is added to the base level B_H in Equation 4, and for the other, it is subtracted. The overall output signals $u_{H1,2}(t)$ were limited to the interval of 0 and 8 kV to avoid premature breakdown of the actuators at higher voltages and prevent force generation at reversed polarity, i.e., $u_H(t) < 0$.

Reproduction and suppression of tremors in the mechanical patient

We developed a custom scale to rate tremor severity (Table S2) in accordance with clinically validated tremor rating scales: the modified Fahn-Tolosa-Marin scale, which was developed by the Mayo Clinic in Rochester (Minnesota, USA),[63] and the Movement Disorder Society - Unified Parkinson's Disease Rating Scale, Part III.[64] We quantified the amplitude of each grade of tremor as listed in Table 1. To generate tremors of different severity grades according to the custom scale, we tuned the input pressure to the PAMs at a given frequency. We often observed some transient behavior of the PAMs and therefore discarded the first 60–120 s of recordings after turning the open-loop tremor generation on in all experiments. We recorded the wrist angle from the encoder for 120 s to characterize the performance and stability of the PAMs at each pressure and frequency setting. With this method, we were able to generate all grades between 2 and 4 Hz, grades 1–3 at 6 Hz, grades 1 and 2 at 8 Hz, and only grade 1 at 10 Hz.

After attaching the Peano-HASEL actuators to the mechanical patient, we adjusted the PAM pressures to fit with the tremor rating scale. We then activated the PID controller of the HASELs to tune the parameters of the PID controller. The proportional (*P*) and integral (*I*) parameters were set to 5 and 1, respectively, while the derivative (*D*) was manually adjusted between 0.08 and 0.11 for maximal suppression at each tremor grade and frequency.

The final recording of the difference between suppressed and unsuppressed conditions was done by generating a tremor for 60 s and then activating the PID controller for the Peano-HASELs for 20 s, followed by another 20 s of inactivated amplitude recording. This cycle was repeated for a total of six times for each set of frequency and tremor grade to ensure repeatability. The actuators' stability over time was demonstrated by recording the amplitude over the course of 30 min during these repeated activation and deactivation conditions for one set of frequency (2 Hz) and tremor grade (grade 2).

To quantify the amount of suppression, we extracted the peak wrist angle amplitude from the maximum flexion and extension values in every tremor oscillation (findpeaks() function in MATLAB R2022b). We then averaged the results across all six repetitions and used the following equation to determine the percent of suppression:

$$\%\ \text{suppression} = 100 \cdot \left(A_{\text{generated}} - A_{\text{suppressed}}\right) \big/ A_{\text{generated}} \qquad \text{(Equation 6)}$$

where A is the average amplitude of the generated or suppressed tremor.

Reproduction and suppression of patient data in the mechanical patient

To test the tremor-suppression capabilities of Peano-HASEL actuators in a more realistic scenario, we manually screened the data recorded from the patients for episodes that showed onsets, variations in amplitude, and decay, as well as a peak amplitude of more than 1° in a time frame of 2–3 min. We manually selected four exemplary tremor episodes from four different patients, 2 with ET and 2 with PD (Table S1). These four tremor episodes were transferred to the mechanical patient and subsequently suppressed.

We used the joint angle of the recorded tremor episodes. To remove slow, voluntary wrist movements, we high-pass filtered the data using an 8th-order infinite impulse response (IIR) filter, with a passband frequency of 2 Hz and a passband ripple of 0.2 Hz, for a sample rate of 2,000 s^{-1} to have it oscillate around zero. Then, we rectified these data to allow for calculating a moving average over 5 s. By dividing the entire signal by its maximum, we calculated a normalized envelope signal of the patients' tremor episodes. We generated an input signal $P(t)$ (Equations 1 and 2) using this envelope signal while taking into consideration the feasible pressure range of the mechanical patient, respecting a minimum air pressure threshold that is required to inflate the PAMs and a maximum pressure of 2 bar. Additionally, we determined each patient's characteristic tremor frequency, using Fourier transform (Table S1). We reproduced tremor episodes in the mechanical patient using both the characteristic frequency and the rectified joint angle (Figures 4 and S6).

For the actual measurements, we prepended 120 s to the beginning of the experiment, of a constant pressure of 1 bar, similar to what was done in the characterization experiments. These data were then measured once with the Peano-HASEL actuators attached but deactivated and then a second time with the actuators and the PID controller activated.

Biomechanical model and simulations

A tremor was simulated in a biomechanical model of the forearm and writs based on the upper limb model of Saul et al.,[73] modified as described by McFarland et al.[72] We considered 5 Hill-type muscle models[92] that actuate the wrist (see Figure 5A) for generating the tremor. We implemented the model in OpenSim software (SimTK) and performed forward-dynamic simulations.[93]

To generate a tremor, we implemented an open-loop sinusoidal stimulation to the muscles

$$u(t) = A_u \cdot \sin(2\pi \cdot f_u \cdot t + \delta) \qquad \text{(Equation 7)}$$

where A_u is amplitude, f_u is frequency, and δ is the phase shift, which is set to $\delta = 0$ for the wrist flexors (flexor carpi radialis and flexor carpi ulnaris) and $\delta = \pi$ for the wrist extensors (extensor carpi radialis longus, extensor carpi radialis brevis, and extensor carpi ulnaris). u(t) was clipped to $0.0001 \le u \le 1$. For every given frequency of the investigated tremor range f = [2, 4, 6, 8, and 10] Hz, the stimulation amplitude A_u was heuristically selected to reproduce wrist amplitudes of θ = [1, 2, 4, 8, and 12]°. This corresponds to tremor amplitudes A = [0.3, 0.7, 1.4, 2.7, and 4.0] cm when calculated with an average male hand size of r = 19.3 cm, which, according to the custom rating scale, correlates to grades [1, 1, 2, 3, and 4].

To actively suppress the tremor in the model, we added a sinusoidal joint torque to the wrist joint:

$$\tau(t) = A_\tau \cdot \sin(2\pi \cdot f_\tau \cdot t + \phi_\tau) \qquad \text{(Equation 8)}$$

The frequency of the joint torque f_τ was always identical to the muscle tremor frequency ($f_\tau = f_u$). The torque amplitude A_τ and phase shift ϕ_τ were tuned to minimize the hand tremor amplitude (maximum tremor suppression). The joint torque amplitude A_τ required to maximally suppress the tremor in the model is our estimate for the required assistive torque and was compared to the forces applied by the Peano-HASEL actuators in the hardware experiments.

Electrical safety considerations for Peano-HASEL actuators

We evaluated the electrical safety considerations for Peano-HASEL actuators of the dimensions and materials mentioned in this paper in wearable applications based on an analysis previously published by Sanchez et al.[94] The maximum energy stored in the actuators was calculated using the formula for a parallel-plate capacitor:

$$0.5 \cdot CV^2 \qquad \text{(Equation 9)}$$

where V is the applied voltage and C is the maximum capacitance, determined as

$$C = \varepsilon_0 \varepsilon_r \cdot A/d \quad \text{(Equation 10)}$$

where ε_0 is the permittivity of free space. For our design and materials system, the permittivity of the film is $\varepsilon_r = 3.3$, the area of the electrode is $A = 0.042\ m^2$, and the distance d between the electrodes when they are fully zipped is equal to twice the film thickness $d = 40\ \mu m$.

Using these parameters, the maximum capacitance of a single Peano-HASEL actuator is 3.07 nF. At the maximum operating voltage of 8 kV, the stored energy is 98.2 mJ, well below the 250-mJ safety threshold. Additionally, the current was limited to 2 mA, below the 5-mA safety threshold specified in IEC 60479-1.

RESOURCE AVAILABILITY

Lead contact

Further information for resources and data will be fulfilled by the lead contact, Daniel Haeufle (daniel.haeufle@uni-tuebingen.de).

Materials availability

This study did not generate new materials.

Data and code availability

All data are available in the main text, supplemental information, and the following repository at FDat, University of Tuebingen: https://doi.org/10.57754/FDAT.g6r73-02j63.

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AUTHOR CONTRIBUTIONS

Conceptualization, A.S., D.H., S.S., and C.K.; patient data collection and analysis, J.S.-T., I. Wurster, P.K., D.W., and D.H.; mechanical patient design and integration, T.N., A.B., and E.H.R.; Simulink environment, T.N., A.B., Y.S., J.R.W., C.C.-M., and D.H.; HASELs fabrication and evaluation, A.S., C.C.-M., E.H.R., and J.R.W.; mechanical patient data collection and analysis, A.S. and C.C.-M.; simulation, J.S.-T., I. Wochner, and D.H.; visualization, A.S., C.C.-M., J.R.W., and J.S.-T.; supervision, D.H., S.S., and C.K.; writing – original draft, A.S., C.C.-M., D.H., J.S.-T., and J.R.W.; writing – review & editing, all authors.

DECLARATION OF INTERESTS

C.K. is a coinventor on three patents, which cover the fundamentals and basic designs of HASEL actuators (assignee of all three patents is "The Regents of the University of Colorado": US patent 10995779B2, granted 2021-05-04; US patent 11486421B2, granted 2022-11-01; and US patent 11408452B2, granted 2022-08-09). C.K. is a cofounder of Artimus Robotics, a start-up company that commercializes HASEL actuators.

SUPPLEMENTAL INFORMATION

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